ASHUTOSH SHARMA

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Generative AI Engineer

PROFESSIONAL SUMMARY

Generative AI Engineer with 3.5+ years building production RAG platforms, agentic LLM systems, and ML pipelines. Progressed from ML analytics to leading enterprise Agentic AI development architecting multi-node workflows that lifted answer faithfulness 40% to 80%+ and cut operational cost 50%. Delivered scalable AI solutions processing 10,000+ queries monthly with 98% accuracy

KEY HIGHLIGHTS

- **80%+ answer faithfulness** on enterprise RAG via NLI citation verification + CRAG filtering, cutting hallucinations 60%
- **6x ingestion format coverage** — added JSON, XLSX, CSV, DOCX, and HTML parsers on top of PDF baseline
- **Sub-second p95 latency** on streaming RAG service over 5,000+ daily requests on FastAPI + Kubernetes
- **92% tool-selection accuracy** on retrieval tools with loop prevention

TECHNICAL SKILLS

Generative AI & LLM: Llama Index, Lang Chain, RAG, Agentic AI, React Agents, Prompt Engineering, RAGAS, CRAG, NLI citation verification, Hugging Face, Transformers

AI/ML: NLP, Deep Learning, Topic Modeling, Classification, Clustering, Feature Engineering, scikit-learn, PyTorch, Random Forest, XGBoost, Pandas, NumPy, Matplotlib

Backend & Data: Python, FastAPI, PostgreSQL, PGvector, SQL, WebSocket, REST APIs, Vector Databases

Cloud, DevOps: Docker, Kubernetes, Helm, Git, CI/CD, Phoenix Tracing, AWS, Object Storage (S3)

Soft Skills: Communication, Teamwork, Cross-Functional Collaboration

PROFESSIONAL EXPERIENCE

Data Scientist | T-Systems ICT India Pvt. Ltd.

December 2022 – Present

AI Operator – Agentic RAG Platform | *May 2025 – Present*

- Architected a multi-node RAG platform on FastAPI, Llama Index, and PostgreSQL, lowering query latency 35% under production load and designed query decomposition, React search, synthesis, and citation nodes as composable Llama Index workflow steps
- Engineered React Agent across 6 specialized retrieval tools, delivering 92% tool-selection accuracy with loop-prevention safeguards. Integrated NLI-based citation verification and CRAG filtering, raising faithfulness from 40% to 80%+ and cutting hallucinations 60%
- Extended ingestion pipeline to support JSON, XLSX, CSV, DOCX, and HTML, onboarding 15+ enterprise data sources. Slashed manual preprocessing 65% through programmatic schema normalization and structured table extraction
- Build evaluation pipeline using RAGAS metrics for failure analysis and prompt optimization, enabling 15% faster iteration cycles across 1,000+ weekly evaluations

Insurance Healthcare Agent – Agentic RAG System | *December 2024 – April 2025*

- Built Agentic RAG system integrating document and PostgreSQL sources, increasing query resolution efficiency by 40% and customer satisfaction by 35%
- Optimized agent workflows using LLM-powered solutions, reducing agent workload by 50% and cutting customer wait times by 70%
- Deployed scalable architecture processing 10,000+ monthly queries with 98% response accuracy and 99.5% system uptime

Crew AI Multi-Agent Content System | *July 2024 – November 2024*

- Orchestrated a modular crew AI multi-agent framework for scalable content generation across 5 distinct use cases
- Cut content production time by 80% on the blog creation pipeline while maintaining editorial quality standards

Topic Modeling for Document Clustering | *February 2024 – July 2024*

- Analyzed 1,000+ enterprise documents using LDA topic modeling, boosting search and content organization by 45%
- Trained a Light GBM classifier reaching 90% accuracy on extracted topics, eliminating 70% of manual categorization effort

Enterprise Data Analytics & ML Modeling | *December 2022 – January 2024*

- Designed customer segmentation and classification models using XG Boost and Random Forest, reaching 87% prediction accuracy
- Streamlined ETL pipelines processing 50K+ records with feature engineering, compressing data preparation time 60%
- Delivered interactive analytics dashboards and reporting pipelines for enterprise clients, improving decision cycles 30%

Intern Data Scientist | Analytics Labs

November 2021 – November 2022

- Constructed a Random Forest model for customer attrition analysis, driving a 6-basis point increase in monthly retention and a 12% reduction in targeted segment attrition
- Established Key Performance Indicators in the Memory Evaluation Matrix, enabling real-time CPE memory tracking across 500+ devices

EDUCATION

Master of Science in Data Science | Manipal Academy of Higher Education, Manipal, India | July 2022 – July 2024

Bachelor of Science in Computer Science, Mathematics & Statistics | Bangalore North University, Bangalore, India | July 2018 – *November* 2021

CERTIFICATIONS

- Generative AI with Large Language Models –Deep Learning. AI / Coursera
- Lang Chain for LLM Application Development –Deep Learning. AI
- Machine Learning Specialization – Coursera (Andrew Ng)

LANGUAGES

English – Professional Working Proficiency | Hindi – Native